Multi-person Physics-based Pose Estimation for Combat Sports

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Pipeline

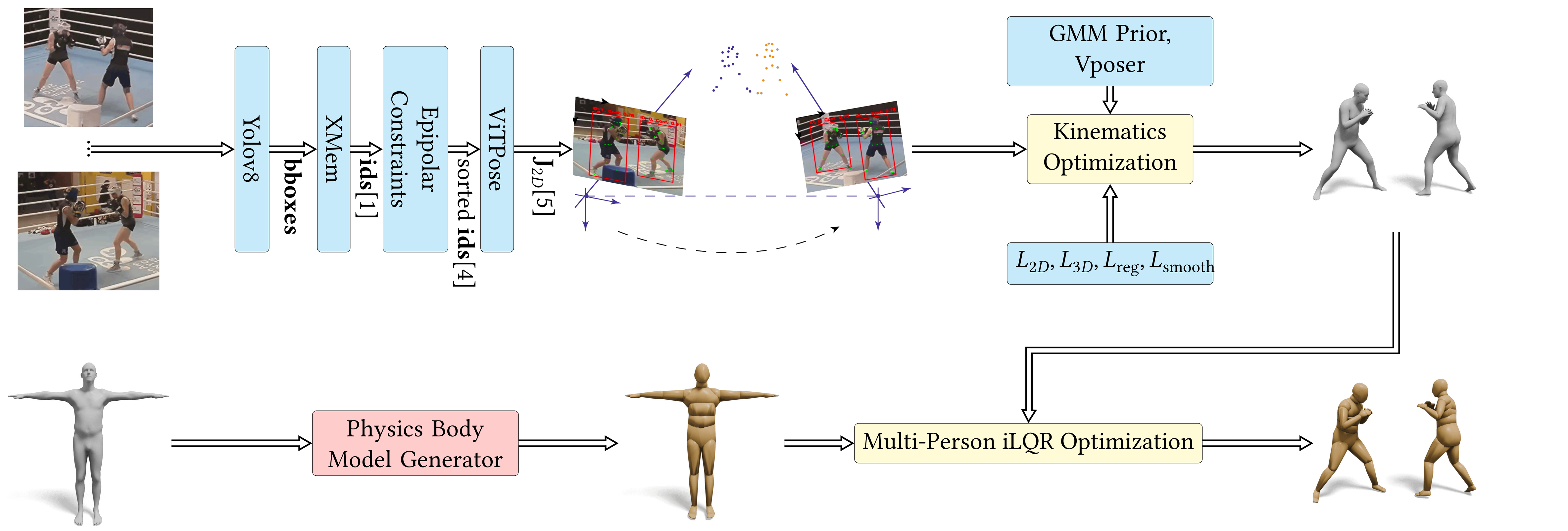


Figure 1: Overview of our proposed pipeline. Initially, bounding boxes (**bboxes**) and robust identity tracking (id) are generated for each individual in the scene. These tracking results facilitate accurate 2D pose estimation ($\mathbf{J}_{2D}$) using ViTPose [5]. The subsequent triangulation yields smoothed 3D keypoints ($\mathbf{J}_{3D}$). These keypoints feed the kinematic optimisation stage, which outputs SMPL parameters (θ, β). Finally, the optimised 3D joint positions (**p**), pose states (**q**), and velocity states (**v**) guide an iLQR-based model-predictive controller [2] to remove physical artefacts and refine motion quality.

Weighted Triangulation and Filtering

Weighted triangulation solves

$$\mathbf{A}\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0, \qquad \mathbf{A} = \begin{bmatrix} \mu_1(P_{11} - u_1 P_{31}) \\ \mu_1(P_{21} - v_1 P_{31}) \\ \vdots \\ \mu_N(P_{1N} - u_N P_{31N}) \\ \mu_N(P_{2N} - v_N P_{32N}) \end{bmatrix}.$$

SVD yields (x, y, z). Cubic-spline smoothing removes outliers; an EKF fills missing frames when triangulation fails.

Kinematics Optimization

Our kinematics optimization refines the SMPL model to minimize the difference between the provided 2D poses ($\mathbf{J}_{2D}$) from multiple views and 3D keypoints ($\mathbf{J}_{3D}$) obtained from triangulation. The process ensures temporal coherency and natural motion by incorporating smoothness and human motion priors. An LBFGS optimizer is used to solve the minimization problem, considering various loss terms, including 2D re-projection, 3D alignment, smoothness, and prior losses.

Kinematics Optimization Formulation: $\min_{\theta} w_1 \mathrm{L}_{2D} + w_2 \mathrm{L}_{3D} + w_3 \mathrm{L}_{reg} + w_4 \mathrm{L}_{smooth} + w_5 \mathrm{L}_{GMM} + w_6 \mathrm{L}_{Vposer}$

- **2D Re-projection Loss**: Aligns the 3D joints with 2D joints using re-projection, weighted by confidence.
  $\mathrm{L}_{2D} = \sum_{j \in \mathcal{V}} \sum_{i \in \mathbf{J}_{2D}} c_{j,i} \rho(\mathrm{J}_{\mathrm{proj}_{j,i}} - \mathrm{J}_{2D_{j,i}})$
- **3D Alignment Loss**: Minimizes the difference between predicted and actual 3D joint positions considering confidence.
  $\mathrm{L}_{3D} = \sum_{i \in \mathbf{J}_{3D}} c_i \|\mathrm{J}_i(\theta, \beta) - \mathrm{J}_{3D,i}\|^2$
- **Smoothness Loss**: Ensures temporal consistency by minimizing differences in poses and vertices over time.
  $\mathrm{L}_{smooth} = \sum_t \|\theta^t - \theta^{t-1}\|^2 + \|\mathcal{M}(\theta^t, \beta) - \mathcal{M}(\theta^{t-1}, \beta)\|^2$
- **Prior Loss**: Guides the optimization towards natural poses using GMM and Vposer priors.
  $\mathrm{L}_{GMM} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{GMM}(\theta_i, \beta), \quad \mathrm{L}_{Vposer} = \frac{1}{N}\sum_{i=1}^{N}(z(\theta_i)^2)$

Physics Body Model Generator

Our goal is to reconstruct the motion of K physics-based humanoids at each time step t. The combined state is $\mathbf{x}_t = (\mathbf{q}_t, \mathbf{v}_t)$, where $\mathbf{q}_t \in \mathbb{R}^{63K}$ is the joint rotations vector, and $\mathbf{v}_t \in \mathbb{R}^{62K}$ is the joint velocities vector. The 3D SMPL joint positions are $\mathbf{p}_t \in \mathbb{R}^{21K}$. The vector $\mathbf{q}_t$ uses minimal coordinates for joint rotations, while $\mathbf{p}_t$ represents the 3D positions relative to the humanoid's body frame, which are used in optimization. We used the Mujoco [3] XML format for multi-humanoid modeling.

Multi-person Dynamics Optimization

The iLQR algorithm [2] optimizes a control trajectory $\mathbf{u}_{0:T}$ by minimizing the cost function. The reference positions. $\mathbf{p}_t$ and velocities $\mathbf{v}_t$ are extracted from the kinematics optimization at each frame t, and the trajectory optimization is "warm-started" using control parameters $\mathbf{u}_{t-1}$ from the previous frame. The control is updated iteratively as $\Delta\mathbf{u}_t = \mathbf{K}_t\Delta\mathbf{x}_t + \alpha\mathbf{k}_t$, with $\mathbf{K}_t$ and $\mathbf{k}_t$ refined until convergence.

Dynamics Optimization Formulation: $\min_{\mathbf{u}_{0:T}} \sum_{t \in [0,T]} w_1 \mathrm{L}_{reg,t} + w_2 \mathrm{L}_{p,t} + w_3 \mathrm{L}_{v,t} + w_4 \mathrm{L}_{collision,t}$

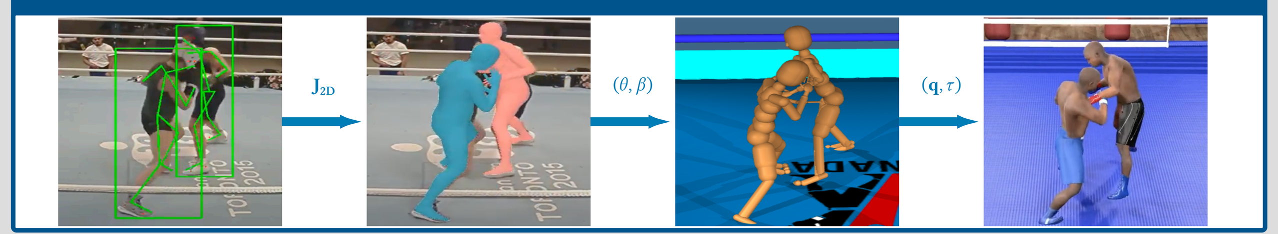
- **Position Loss**: Minimizes the difference between predicted and actual 3D joint positions in each short horizon.
  $\mathrm{L}_{p,t} = \sum_{i \in \mathbf{J}_{3D}} \|\mathrm{p}_{i,t} - \mathrm{J}_{3D,i,t}\|^2$
- **Velocity Loss**: Minimizes the difference between predicted and actual 3D joint velocities in each short horizon.
  $\mathrm{L}_{v,t} = \sum_{i \in \mathbf{J}_{3D}} c_i \|\mathrm{v}_{i,t} - \mathrm{v}_{3D,i,t}\|^2$
- **Collision Loss**: The collision term penalizes the penetration between the geoms of different humanoids.
  $\mathrm{L}_{collision,t} = \sum_{i \in 1 \ldots M} \sum_{j \in 1 \ldots M} \max(0, \epsilon - \mathrm{dist}(G_{1_i}, G_{2_j}))$
- **Regularization Loss**: Prevent sudden changes in control input. $\mathrm{L}_{reg,t} = \|\mathbf{u}_t\|^2$

Discussion and Future Works

We created a 10-hour dataset of boxing motions to develop a model for generating intelligent VR responses.

Video

Visualization



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